

AI Network Security Analysis Report

IoT_Botnet_DDoS

Generated: 2025-08-03 22:08:56

Executive Summary

The AI-powered network security monitoring system analyzed a IoT_Botnet_DDoS scenario involving 1,074 network packets over a 0.1 minute period. The system detected 0 security threats with an overall accuracy of 0.0%.

Attack Narrative

Phase 1: IoT Device Discovery

Duration: 10.0 minutes

Packets Analyzed: 8

Threat Level: low

Automated scanning for vulnerable IoT devices with default credentials

AI Detection: Normal activity detected during IoT Device Discovery. Continuing monitoring.

Recommended Actions: Continue monitoring

Phase 2: Mass Device Compromise

Duration: 20.0 minutes

Packets Analyzed: 8

Threat Level: low

Exploitation of IoT devices using known vulnerabilities and default passwords

AI Detection: Normal activity detected during Mass Device Compromise. Continuing monitoring.

Recommended Actions: Continue monitoring

Phase 3: Botnet Formation

Duration: 15.0 minutes
Packets Analyzed: 8
Threat Level: low

Infected devices connect to command & control server and await instructions

AI Detection: Normal activity detected during Botnet Formation. Continuing monitoring.
Recommended Actions: Continue monitoring

Phase 4: DDoS Attack Launch

Duration: 30.0 minutes
Packets Analyzed: 1000
Threat Level: low

Coordinated DDoS attack from thousands of compromised IoT devices

AI Detection: Normal activity detected during DDoS Attack Launch. Continuing monitoring.
Recommended Actions: Continue monitoring

Phase 5: Attack Persistence

Duration: 15.0 minutes
Packets Analyzed: 50
Threat Level: low

Botnet maintains attack while evading detection and mitigation attempts

AI Detection: Normal activity detected during Attack Persistence. Continuing monitoring.
Recommended Actions: Continue monitoring

Machine Learning Insights

- ■ ML detected anomaly (score: 0.1)
- ■ Network health optimal (93%)
- ■ ML detected anomaly (score: 0.0)
- ■ Top talker: external_scanner (300 packets)
- ■ Top talker: external_scanner (293 packets)
- ■ Top talker: external_scanner (253 packets)
- ■ Top talker: external_scanner (248 packets)
- ■ Top talker: external_scanner (247 packets)
- ■ Top talker: external_scanner (196 packets)
- ■ Top talker: external_scanner (188 packets)

Critical Findings

- 1. **Most Affected Devices:**
- 2. **Primary Attack Vectors:** none
- 3. **Data Exfiltration Risk:** Low
- 4. **Network Integrity Status:** Partially Compromised

Response Effectiveness

Metric	Value
Mean Time to Detect (MTTD)	0.00 seconds
Detection Rate	0.0%
False Positive Rate	0.0%
Overall Accuracy	0.0%

Security Recommendations

- 1. Enhance detection capabilities with additional ML training
- 2. Conduct security awareness training for all users
- 3. Regular penetration testing and vulnerability assessments
- 4. Enable multi-factor authentication across all systems

5. Implement Zero Trust security model